

TOWARDS UNBIASED ON-POLICY DISTILLATION FOR BLOCK DIFFUSION LANGUAGE MODELS

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ABSTRACT

On-policy distillation (OPD) has emerged as an effective post-training paradigm for language models, with recent efforts extending it to block diffusion language models (BDLMs). However, existing studies focus almost exclusively on small block sizes, leaving distillation into student models with larger blocks underexplored. In this work, we investigate this regime and reveal two critical optimization biases that induce severe training instability. First, mismatched block boundaries between teacher and student cause *context misalignment*, providing distorted supervisory signals that misguide student decoding. Second, even under aligned contexts, an *intrinsic optimization bias* in OPD, where the student tends to rapidly absorb high-support signals while lagging on low-support updates, drives a premature confidence surge that traps weaker students in catastrophic overconfidence collapse. To resolve these, we propose **Un-OPD**, an unbiased on-policy distillation framework with two novelties for stabilizing BDLM training. First, Un-OPD introduces a boundary-aware step filtering strategy that eliminates context-misaligned decoding steps. Second, Un-OPD proposes moderating optimization intensity at high-support positions via a support-rebalanced confidence calibration, thereby bypassing overconfidence collapse. Beyond stability, we also introduce a rollout reuse mechanism to reduce rollout generation overhead. Extensive experiments on math reasoning and code generation benchmarks show that Un-OPD consistently stabilizes training and delivers superior performance while reducing wall-clock training time by approximately half.

1 INTRODUCTION

Diffusion large language models (dLLMs) have recently emerged as a compelling alternative to autoregressive (AR) models (Achiam et al., 2023; Hui et al., 2024; Yang et al., 2025) by generating multiple tokens in parallel through iterative denoising (Nie et al., 2025; Ye et al., 2025). For example, some open-source models like LLaDA (Nie et al., 2025) and Dream (Ye et al., 2025) maintain a long sequence of mask tokens and iteratively reveal tokens through parallel denoising steps. To inherit the advantages of AR models, particularly KV caching compatibility (Kwon et al., 2023; Ni et al., 2026) and flexible sequence extension, recent dLLM works have advanced toward block diffusion large language models (BDLMs) (Arriola et al., 2025; Cheng et al., 2026; Tian et al., 2025; Wu et al., 2026b; Bie et al., 2025; 2026). By decomposing a sequence into consecutive blocks, BDLMs perform global reasoning autoregressively across blocks while generating tokens via parallel diffusion within each block, achieving a superior quality-latency trade-off.

Following the success of post-training in AR models, particularly reinforcement learning with verifiable rewards (RLVR) (Shao et al., 2024; Yu et al., 2026), reinforcement learning (RL) has been actively adapted to enhance the reasoning capabilities of dLLMs (Pan et al., 2026a; Zhao et al., 2026; Wang et al., 2026b; Ou et al., 2026; Ma et al., 2026; Ni et al., 2026). Specifically, to bypass the intractable conditional likelihood estimation inherent in diffusion denoising, pioneering efforts introduce mean-field surrogates (Zhao et al., 2026; Tang et al., 2026) or tree-structured policy updates (Pan et al., 2026a). Nevertheless, these approaches either rely on coarse outcome-based

rewards leading to inaccurate and sparse credit assignment across multi-step denoising trajectories (Shao et al., 2026; Zhao et al., 2026), or incur substantial inference overhead to acquire finer-grained reward signals (Pan et al., 2026a). Recently, on-policy distillation (OPD) (Hou et al., 2026; Yang et al., 2026b) has emerged as another promising post-training paradigm. By aligning the student with teacher predictions along its own rollout trajectories, OPD provides dense, token-level supervision. Recent studies (Hou et al., 2026; Yang et al., 2026b; Fu et al., 2026) demonstrate that OPD outperforms previous RLVR algorithms on language models in terms of both optimization stability and sample efficiency.

Despite extensive progress in AR models, OPD for dLLMs remains at an early stage. dOPSD (Dat et al., 2026) extends self-distillation to full-sequence denoising models such as LLaDA (Nie et al., 2025) using predictions from later states with more revealed tokens to supervise earlier states. This design is less applicable to BDLMs, where later states reveal context only within the current block. TOPD (Ren et al., 2026) extends OPD to BDLMs by distilling along the student’s denoising trajectories, but it focuses exclusively on small block sizes, leaving distillation into students with larger blocks, which offer inherently greater parallel decoding potential, largely underexplored. In this work, we investigate scaling OPD to BDLMs with larger block sizes. We use small-block BDLMs as teachers because they provide higher-quality generations, and distill them into students with varying block sizes.

Our work uncovers two underlying optimization biases that impede the effective distillation of BDLMs: **(1) Context misalignment.** When denoising within a larger block, the student receives biased supervision because the teacher cannot access its authentic decoding states across mismatched boundaries. Such signals tend to be overly aggressive or excessively conservative, disrupting the student’s confidence-driven decisions. **(2) Biased learning dynamics.** During OPD, the student prioritizes tokens with strong teacher support over weakly supported ones, causing confidence to prematurely surge and then fall back. This spurious surge drives fragile students into low-quality decoding regimes that are difficult to rectify. To address these two problems, we propose **Un-OPD** (Unbiased On-Policy Distillation), a simple framework for scaling OPD to BDLMs with larger block sizes. Specifically, we first restrict OPD to decoding states with context alignment, ensuring impartial teacher supervision signals. We then quantify *teacher support* to distinguish strongly supported positions and rebalance the teacher support by moderating the update intensity, preventing over-confidence collapse. Beyond stability, we further introduce a *rollout reuse* mechanism that reuses sampled trajectories for reducing rollout generation overhead.

In summary, the main contributions of this work are as follows:

- We pioneer the study of OPD on BDLMs with varying block sizes, identifying two underlying optimization bottlenecks: *context misalignment* arising from mismatched block sizes and *biased learning dynamics* inherent to standard OPD.
- We propose **Un-OPD**, an unbiased OPD framework that integrates context-aligned step selection and support-rebalanced confidence calibration to stabilize the training dynamics. We also introduce a rollout reuse mechanism to reduce rollout overhead during training.
- We conduct a comprehensive evaluation across various block sizes and demonstrate that Un-OPD achieves superior performance over counterparts, alongside greater training efficiency.

2 RELATED WORK

Block Diffusion Language Models. dLLMs (Nie et al., 2025; Zhu et al., 2025; Ye et al., 2025) offer an alternative to autoregressive LLMs (Yang et al., 2025; Achiam et al., 2023) through iterative discrete denoising, with recent gains from scaling data and model size. LLaDA (Nie et al., 2025; Zhu et al., 2025) trains from scratch via masking and unmasking, whereas Dream (Ye et al., 2025) adapts pretrained autoregressive models. Full-sequence dLLMs, however, struggle to combine coherent chain-of-thought reasoning with efficient inference mechanisms such as KV caching (Kwon et al., 2023; Ni et al., 2026). Block diffusion language models (BDLMs) (Arriola et al., 2025; Cheng et al., 2026; Tian et al., 2025; Wu et al., 2026b; Bie et al., 2025; 2026) address this trade-off by generating blocks autoregressively and denoising tokens in parallel within each block. Typically adapted from autoregressive models, BDLMs aim to retain their reasoning capabilities while enabling parallel intra-block generation.

On-Policy Distillation. By aligning student-generated rollouts with per-token teacher predictions, on-policy distillation (OPD) (Hou et al., 2026; Yang et al., 2026b; Gu et al., 2026; Fu et al., 2026; Xie et al., 2026; Xu et al., 2026; Zhang et al., 2026; Jin et al., 2026; Wang et al., 2026a; Xing et al., 2026) has demonstrated superior post-training efficiency compared to GRPO-style optimization (Shao et al., 2024; Yu et al., 2026; Gao et al., 2025; Tan et al., 2025) and off-policy counterparts (Wang et al., 2023; Ren & Sutherland, 2025; Wu et al., 2026a). To extract more reliable teacher supervision, several works (Zhang et al., 2026; Wang et al., 2026a; Xu et al., 2026; Xie et al., 2026; Zheng et al., 2026) focus optimization on student rollout prefixes. In parallel, on-policy self-distillation (OPSD) (Lu et al., 2026; Yang et al., 2026a; Pan et al., 2026b; Yuan et al., 2026) uses the same model as the teacher, conditioned on privileged context such as ground-truth answers. TOPD (Ren et al., 2026) extends OPD to BDLMs (Cheng et al., 2026) with small block sizes. We investigate instability at larger student block sizes and address context misalignment and biased optimization dynamics.

Efficient RLVR via Rollout Replay. To improve data efficiency in RLVR, recent works incorporate rollout replay into GRPO-style optimization, curbing policy drift via on/off-policy interleaving (Zhang et al., 2025; Zhan et al., 2026; Li et al., 2025), quality filtering (Zhang et al., 2025; Wang et al., 2025), or importance-sampling corrections (Mao et al., 2026). However, these methods require delicate heuristics to manage staleness and off-policy bias. In contrast, we extend rollout replay to OPD, which exhibits fundamentally higher tolerance to trajectory staleness. This allows Un-OPD to employ an aggressive replay schedule that substantially reduces rollout generation overhead without the complexity of trace filtering or data mixing.

3 METHOD

We propose **Un-OPD**, a framework that scales OPD to BDLMs with large block sizes by integrating two recipes: context-aligned distillation and support-rebalanced distillation. Our design is driven by two key sources of training instability: *context misalignment* arising from mismatched block boundaries and *biased learning dynamics* inherent to standard OPD. In this section, we first provide the preliminaries in Sec. 3.1. Next, Sec. 3.2 analyzes context misalignment and introduces context-aligned distillation. Then, Sec. 3.3 investigates the biased learning dynamics of OPD and introduces our support rebalance strategy. Finally, Sec. 3.4 unifies these components into the overall Un-OPD training framework while introducing rollout reuse to reduce trajectory-sampling overhead.

3.1 PRELIMINARIES

On-policy distillation. On-policy distillation (OPD) trains a student policy π_θ on its own rollouts using token-level supervision from a fixed teacher policy π_T . Given a prompt $q \sim \mathcal{D}$, the student generates a response $\tau = (o_1, \dots, o_{|\tau|})$ where $|\tau|$ is the length of the trajectory. Let c_i denote the conditioning context at which token o_i is predicted, including the prompt and the visible decoding state. In the reverse-KL formulation considered here, the student minimizes

$$\mathcal{L}_{\text{OPD}}(\theta) = \mathbb{E}_{q \sim \mathcal{D}, \tau \sim \pi_\theta(\cdot | q)} \left[\frac{1}{|\tau|} \sum_{i=1}^{|\tau|} D_{\text{KL}}(\pi_\theta(\cdot | c_i) \parallel \pi_T(\cdot | c_i)) \right], \quad (1)$$

where sampled rollout contexts are held fixed during each optimization update. The reverse KL at position i is computed over the vocabulary $\mathcal{V}$:

$$\mathcal{L}_i = \sum_{v \in \mathcal{V}} \pi_\theta(v | c_i) [\log \pi_\theta(v | c_i) - \log \pi_T(v | c_i)]. \quad (2)$$

Trace-aligned OPD for block diffusion language models (BDLMs). BDLMs generate sequences in a semi-autoregressive fashion: they proceed block by block, iteratively revealing tokens within the current block while conditioning on all preceding, fully generated blocks. Following TOPD (Ren et al., 2026) and TraceRL (Wang et al., 2026b) (a GRPO-based method for block diffusion language models), we tailor OPD to this decoding paradigm by supervising the student directly along its on-policy denoising trajectory. Specifically, given a prompt q , the student samples an execution trace $\xi = (s_0, a_0, s_1, \dots, a_{K-1}, s_K)$, where s_k denotes the intermediate decoding state (with partially

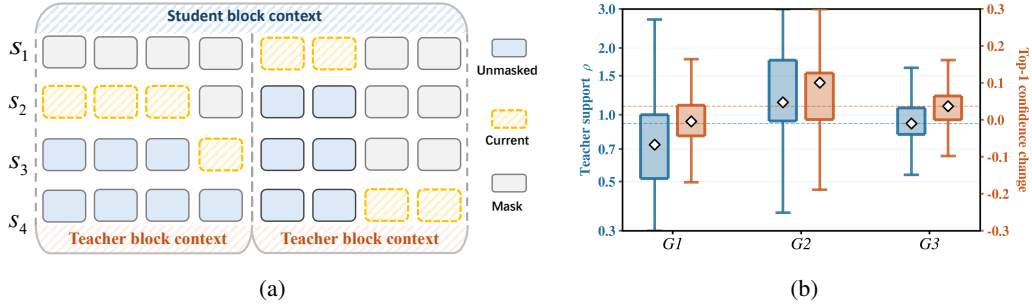


Fig. 1: **Context misalignment and its effects on OPD.** (a) The student denoises an 8-token block, while the teacher provides supervision within 4-token blocks. Contexts are misaligned at states s_1-s_3 and aligned at s_4 . (b) Box plots show distributions of teacher support ρ (blue) and post-OPD top-1 probability change (orange) for context regimes G_1, G_2 , and G_3 . The diamonds mark mean values.

unmasked tokens), $a_k = \{(i, x_i)\}$ represents the set of token decisions committed at step k , and s_K is the finalized sequence. For each committed position $(i, x_i) \in a_k$, we evaluate $\pi_T(\cdot \mid q, s_k, i)$ and $\pi_\theta(\cdot \mid q, s_k, i)$ and minimize their reverse-KL divergence.

3.2 CONTEXT ALIGNED DISTILLATION

Understanding context misalignment. Unlike autoregressive teacher–student pairs with identical causal contexts, BDLMs with different block sizes can have misaligned visible contexts even under trace-aligned OPD. Fig. 1(a) illustrates this mismatch for teacher and student block sizes of 4 and 8. The student undergoes four commit steps within its 8-token block (yielding states s_1-s_4), whereas the teacher’s span is split into two 4-token blocks by the red dashed boundary. Consequently, their contexts align only at s_4 , while s_1-s_3 suffer from distinct mismatches: at s_1 , the teacher conditions on the fully generated left-block context that remains noisy to the student, whereas at s_2 and s_3 , the student accesses unmasked context from the right-block that lies strictly beyond the teacher’s first block boundary.

Optimization bias under context misalignment. To probe how context misalignment affects student optimization, we pair an SDAR-4B-b8 student with a TraDo-8B-b4 teacher and categorize decoding states into three regimes: G_1 (student sees more context such as s_2, s_3), G_2 (teacher sees more such as s_1), and G_3 (contexts are aligned such as s_4). Let $h_i = \arg \max_v p_{S,i}(v)$ denote the student’s top-1 token at position i . We quantify the teacher support $\rho_i = p_{T,i}(h_i)/p_{S,i}(h_i)$, where $\rho_i > 1$ and $\rho_i < 1$ denote stronger and weaker teacher support for the student’s top-1 prediction, respectively. In Fig. 1(b), the blue box plots show the teacher support distributions across the three state groups, while the orange box plots show the distributions of changes in the student’s top-1 probability after training with OPD exclusively on each group. The teacher support skews predominantly below 1.0 in G_1 and above 1.0 in G_2 , whereas G_3 remains well-balanced. The confidence change mirrors this pattern: G_1 yields negligible confidence change, G_2 causes a sharp surge, and G_3 produces a moderate rise. These results demonstrate that context asymmetry systematically biases teacher supervision signals on the student rollout tokens in two opposite directions:

- **Under-supportive supervision signals.** With less visible context than the student, the teacher tends to underestimate the student’s valid rollouts relative to aligned contexts, which leaves the student at risk of **failing to reinforce confidence in correct predictions** through OPD.
- **Over-supportive supervision signals.** With more visible context than the student, the teacher tends to provide overly aggressive support relative to aligned contexts, which puts the student at risk of **becoming overconfident in incorrect predictions** through OPD.

Context aligned step selection. Let $s_{T,k}$ and $s_{S,k}$ denote the teacher and student block states at step k . To mitigate context-induced bias, we retain only decisions committed at steps with aligned visible contexts:

$$\mathcal{A}_{\text{align}} = \{(k, i) : (i, x_i) \in a_k, 0 \leq k < K, s_{T,k} = s_{S,k}\}. \quad (3)$$

Here, (k, i) denotes position i committed at step k . Averaging the reverse KL over these positions gives the context-aligned OPD objective:

$$\mathcal{L}_{\text{aligned-OPD}} = \frac{1}{|\mathcal{A}_{\text{align}}|} \sum_{(k, i) \in \mathcal{A}_{\text{align}}} D_{\text{KL}}(\pi_{\theta}(\cdot | q, s_{\text{S}, k}, i) \| \pi_{\text{T}}(\cdot | q, s_{\text{T}, k}, i)). \quad (4)$$

3.3 SUPPORT-REBALANCED DISTILLATION

Though we have identified and mitigated context-induced bias in Sec. 3.2, training remains sub-optimal. Crucially, we observe that another bias, the internal optimization dynamics of OPD, also compromises BDLM training stability. In this part, we first present empirical observations of this behavior in OPD, then provide a theoretical grounding, and finally introduce a simple yet effective solution.

Biased optimization dynamics in OPD. Since top-1 token confidence critically dictates decoding decisions in BDLMs, we begin by examining the evolution of the student’s average top-1 confidence throughout the entire OPD training process. As illustrated in Fig. 2, decoding confidence of the student model exhibits a distinct two-stage pattern throughout training: *a rapid surge during the early phase, followed by a slight decline during later training*. To dissect the underlying mechanism behind this phenomenon, we leverage the previously defined teacher support ρ_i in Sec. 3.2 given its tight correlation with confidence shifts, and divide the supervised positions into two distinct groups: $\mathcal{I}_+ = \{i : \rho_i > 1\}$ and $\mathcal{I}_- = \{i : \rho_i < 1\}$. Empirically, updates from $\mathcal{I}_+$ tend to increase the student’s decoding confidence, whereas updates from $\mathcal{I}_-$ tend to decrease it.

As shown in Fig. 2, we reveal that the fraction of high-support positions steadily decreases, while that of low-support positions rises before eventually plateauing. We hypothesize that this early confidence surge stems from an asymmetric learning dynamic: the student **rapidly internalizes** supervisory signals from high-support positions ($\mathcal{I}_+$), whereas updates from low-support positions ($\mathcal{I}_-$) **suffer from an optimization lag**. We term this learning asymmetry the *intrinsic optimization bias* of OPD. Consequently, as the student’s confidence is prematurely driven up, the rollout policy becomes increasingly prone to sampling tokens beyond the teacher’s support region, which in turn produces more low-support positions which ultimately dominate training and trigger a subsequent pullback in confidence.

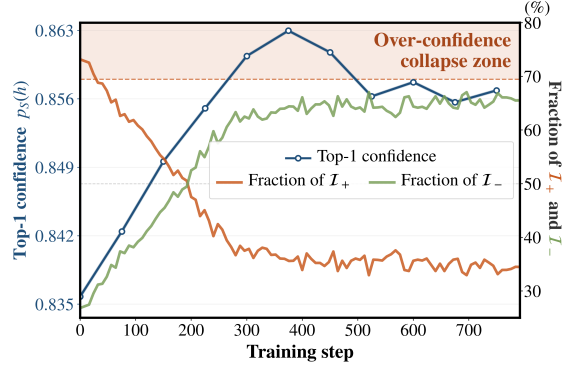


Fig. 2: Evolution of student mean top-1 confidence (blue) alongside subset fractions of $\mathcal{I}_+$ (orange) and $\mathcal{I}_-$ (green). The shaded area denotes the overconfidence collapse zone.

Theoretical grounding. To theoretically characterize this optimization bias, we examine the reverse-KL gradient update to the logit of the student’s top-1 prediction h_i . Let $a_i = p_{\text{S}, i}(h_i)$ and $b_i = p_{\text{T}, i}(h_i)$, such that $\rho_i = b_i/a_i$. We denote by $D_i^{\text{tail}} \geq 0$ the reverse-KL divergence between the renormalized tail distributions of the student and teacher over all non-top-1 tokens ($v \neq h_i$).

Proposition 1 (Asymmetric top-1 logit updates under reverse-KL). *For a fixed context and a teacher distribution with positive token probabilities, a reverse-KL gradient descent step with learning rate $\eta > 0$ directly on the student’s softmax logits yields*

$$\Delta z_{\text{S}, i}(h_i) = \eta a_i (1 - a_i) \left[\log \frac{b_i (1 - a_i)}{a_i (1 - b_i)} + D_i^{\text{tail}} \right]. \quad (5)$$

Proposition 1 shows that the nonnegative tail term D_i^{tail} reinforces upward logit updates for $\rho_i > 1$, but can weaken or reverse downward updates for $\rho_i < 1$. This local logit-level analysis provides a mechanism consistent with the observed asymmetry, although actual training dynamics also depend on the overall training data distribution.

Support-rebalanced confidence calibration. Due to this optimization bias, OPD is not a free lunch for BDLMs. When weaker students are guided by a strong teacher, a premature surge in confidence readily pushes them into a degenerate generation regime (Fig. 2). Consequently, such students slip into a catastrophic overconfidence zone, failing to recover even in later training stages to a confidence regime commensurate with their actual capacity. Here, we propose calibrating the supervision signals specifically at high-support positions. In implementation, we temper the teacher target by interpolating its logits with the student’s detached logits for each $i \in \mathcal{I}_+$, thereby moderating update intensity at high-support positions. Let $\mathcal{I}$ denote the set of supervised token positions and $\lambda \in [0, 1]$ be the teacher interpolation weight at high-support positions. The calibrated teacher target and the resulting reverse-KL objective are defined as:

$$\begin{aligned}\tilde{z}_{T,i}(v) &= [1 - (1 - \lambda)\mathbf{1}_{\{i \in \mathcal{I}_+\}}]z_{T,i}(v) + (1 - \lambda)\mathbf{1}_{\{i \in \mathcal{I}_+\}}\text{sg}[z_{S,i}(v)], \\ \mathcal{L}_\lambda &= \frac{1}{|\mathcal{I}|} \sum_{i \in \mathcal{I}} D_{\text{KL}}(p_{S,i} \parallel \text{softmax}(\tilde{z}_{T,i})).\end{aligned}\tag{6}$$

Here, $\mathbf{1}_{\{i \in \mathcal{I}_+\}}$ indicates a high-support position and $\text{sg}[\cdot]$ denotes stop-gradient. At these positions, the teacher and detached student logits receive weights λ and $1 - \lambda$, respectively; smaller λ yields stronger calibration. Low-support positions retain the original teacher target.

3.4 OVERALL FRAMEWORK

Overall objective. Unifying the two training recipes, we can obtain the unbiased OPD framework for BDLMs, whose overall objective minimizes the reverse-KL divergence over aligned decisions in $\mathcal{A}_{\text{align}}$ using the calibrated teacher target:

$$\mathcal{L}_{\text{Un-OPD}}(\theta) = \mathbb{E}_{q,\xi} \left[\frac{1}{|\mathcal{A}_{\text{align}}|} \sum_{(k,i) \in \mathcal{A}_{\text{align}}} D_{\text{KL}}(p_{S,k,i} \parallel \text{softmax}(\tilde{z}_{T,k,i})) \right], \tag{7}$$

where $p_{S,k,i} = \pi_\theta(\cdot \mid q, s_{S,k}, i)$ and $\tilde{z}_{T,k,i}$ is the calibrated target from Eq. 6. When $\lambda = 1$, Eq. 7 recovers context-aligned OPD (Eq. 4), while $\lambda < 1$ calibrates the teacher target at high-support positions to curb premature confidence inflation.

Rollout reuse. We notice that OPD is inherently better suited for trajectory reuse compared to RLVR. In OPD, historical rollouts serve merely as conditioning states rather than frozen discrete actions or stale scalar rewards as in RLVR. Across multiple optimization steps on the same rollout batch, we recompute the student distributions under the current policy, while the frozen teacher distributions over the fixed states remain unchanged. The calibrated targets and distillation gradients are recomputed accordingly. Consequently, the primary discrepancy in OPD replay arises solely from *state distribution mismatch* between the sampling policy and the updated student. Unlike RLVR, which suffers from volatile importance sampling and reward clipping over historical rollouts, OPD suffers from substantially milder off-policy degradation. This insight motivates our *rollout reuse* mechanism, which performs multiple updates per rollout to effectively reduce the heavy cost of rollout sampling.

4 EXPERIMENTS

In this section, we conduct comprehensive experiments across BDLMs with varying block sizes to evaluate the effectiveness of Un-OPD. We first detail the experimental configurations in Sec. 4.1, then assess how the proposed Un-OPD improves performance across different benchmarks in Sec. 4.2. Finally, we provide an ablation study about the core recipes of Un-OPD in Sec. 4.3.

4.1 EXPERIMENTAL SETTINGS

Baselines and Benchmarks. To comprehensively evaluate the scalability and robustness of Un-OPD, we employ SDAR-4B and SDAR-1.7B (Cheng et al., 2026) with varying block sizes ($b \in \{4, 8\}$) as student BDLMs, paired with a default TRaDo-8B (Wang et al., 2026b) as teacher ($b = 4$). We evaluate the models on standard benchmarks for mathematical reasoning (e.g.,

Tab. 1: Performance comparison on math and code generation benchmarks. We report accuracy (%) with gains over the base model computed by direct subtraction. With the teacher block size fixed at 4, student block sizes of 8 and 4 correspond to **asymmetric** and **symmetric** configurations, respectively. The best results in each group are **boldfaced** and underlined.

Method	Mathematical Reasoning					Code Generation		
	GSM8K	MATH500	AMC2023	Olympiad	Avg.	MBPP	HumanEval	Avg.
<i>Student Model: SDAR-4B-b8</i>								
Base	88.9	66.3	42.5	29.2	56.7	53.5	67.9	60.7
+ SFT	88.1 _{-0.8}	65.4 _{-0.9}	40.3 _{-2.2}	27.9 _{-1.3}	55.4 _{-1.3}	51.7 _{-1.8}	65.5 _{-2.4}	58.6 _{-2.1}
+ TraceRL	89.5 _{+0.6}	69.4 _{+3.1}	46.8 _{+4.3}	30.0 _{+0.8}	58.9 _{+2.2}	54.1 _{+0.6}	68.9 _{+1.0}	61.5 _{+0.8}
+ TOPD	88.4 _{-0.5}	67.1 _{+0.8}	43.6 _{+1.1}	30.8 _{+1.6}	57.5 _{+0.8}	52.3 _{-1.2}	65.1 _{-2.8}	58.7 _{-2.0}
+ Un-OPD (Ours)	90.0 _{+1.1}	70.1 _{+3.8}	45.8 _{+3.3}	32.8 _{+3.6}	59.7 _{+3.0}	55.6 _{+2.1}	71.1 _{+3.2}	63.4 _{+2.7}
<i>Student Model: SDAR-1.7B-b8</i>								
Base	74.8	51.9	25.9	17.3	42.5	41.2	52.4	46.8
+ SFT	73.4 _{-1.4}	50.1 _{-1.8}	23.3 _{-2.6}	15.7 _{-1.6}	40.6 _{-1.9}	40.5 _{-0.7}	53.5 _{+1.1}	47.0 _{+0.2}
+ TraceRL	75.6 _{+0.8}	53.0 _{+1.1}	27.8 _{+1.9}	18.3 _{+1.0}	43.7 _{+1.2}	43.5 _{+2.3}	54.4 _{+2.0}	49.0 _{+2.2}
+ TOPD	74.4 _{-0.4}	50.9 _{-1.0}	25.5 _{-0.4}	17.2 _{-0.1}	42.0 _{-0.5}	40.1 _{-1.1}	50.9 _{-1.5}	45.5 _{-1.3}
+ Un-OPD (Ours)	75.3 _{+0.5}	54.9 _{+3.0}	27.7 _{+1.8}	18.6 _{+1.3}	44.1 _{+1.6}	42.4 _{+1.2}	55.6 _{+3.2}	49.0 _{+2.2}
<i>Student Model: SDAR-4B-b4</i>								
Base	88.1	66.5	43.1	29.5	56.8	50.8	65.2	58.0
+ SFT	89.8 _{+1.7}	67.4 _{+0.9}	40.2 _{-2.9}	31.7 _{+2.2}	57.3 _{+0.5}	54.8 _{+4.0}	71.3 _{+6.1}	63.1 _{+5.1}
+ TraceRL	87.1 _{-1.0}	70.4 _{+3.9}	44.2 _{+1.1}	30.7 _{+1.2}	58.1 _{+1.3}	56.4 _{+5.6}	71.2 _{+6.0}	63.8 _{+5.8}
+ TOPD	89.7 _{+1.6}	70.8 _{+4.3}	45.0 _{+1.9}	32.9 _{+3.4}	59.6 _{+2.8}	56.8 _{+6.0}	70.8 _{+5.6}	63.8 _{+5.8}
+ Un-OPD (Ours)	89.5 _{+1.4}	72.0 _{+5.5}	45.4 _{+2.3}	33.5 _{+4.0}	60.1 _{+3.3}	57.1 _{+6.3}	71.0 _{+5.8}	64.1 _{+6.1}
<i>Student Model: SDAR-1.7B-b4</i>								
Base	74.4	53.2	26.7	16.8	42.8	41.2	53.0	47.1
+ SFT	76.4 _{+2.0}	54.2 _{+1.0}	28.8 _{+2.1}	19.5 _{+2.7}	44.7 _{+1.9}	43.9 _{+2.7}	56.1 _{+3.1}	50.0 _{+2.9}
+ TraceRL	76.7 _{+2.3}	55.9 _{+2.7}	30.6 _{+3.9}	19.0 _{+2.2}	45.6 _{+2.8}	45.4 _{+4.2}	55.9 _{+2.9}	50.7 _{+3.6}
+ TOPD	76.6 _{+2.2}	57.7 _{+4.5}	30.6 _{+3.9}	21.1 _{+4.3}	46.5 _{+3.7}	43.8 _{+2.6}	57.4 _{+4.4}	50.6 _{+3.5}
+ Un-OPD (Ours)	76.1 _{+1.7}	58.8 _{+5.6}	31.5 _{+4.8}	21.6 _{+4.8}	47.0 _{+4.2}	46.1 _{+4.9}	57.9 _{+4.9}	52.0 _{+4.9}

Tab. 2: Ablation of Un-OPD components on SDAR-4B-b8 and SDAR-1.7B-b8. We report accuracy (%) on three mathematical reasoning benchmarks and the corresponding training cost in GPU hours. Components are added cumulatively; subscripts show accuracy changes from the preceding row, computed by direct subtraction.

Method	SDAR-4B-b8				SDAR-1.7B-b8			
	MATH500	AMC2023	Olympiad	GPU hours	MATH500	AMC2023	Olympiad	GPU hours
Base	66.3	42.5	29.2	—	51.9	25.9	17.3	—
+ TOPD	67.1 _{+0.8}	43.6 _{+1.1}	30.8 _{+1.6}	152	50.9 _{-1.0}	25.5 _{-0.4}	17.2 _{-0.1}	104
+ context-aligned	69.5 _{+2.4}	44.5 _{+0.9}	32.6 _{+1.8}	152	51.5 _{+0.6}	26.7 _{+1.2}	17.3 _{+0.1}	104
+ support-rebalanced	70.3 _{+0.8}	45.7 _{+1.2}	33.0 _{+0.4}	152	54.4 _{+2.9}	27.5 _{+0.8}	18.9 _{+1.6}	104
+ rollout reuse	70.1 _{-0.2}	45.8 _{+0.1}	32.8 _{-0.2}	72	54.9 _{+0.5}	27.7 _{+0.2}	18.6 _{-0.3}	48

GSM8K (Cobbe et al., 2021), MATH-500 (Hendrycks et al., 2021), AMC 2023), as well as HumanEval (Chen et al., 2021) and MBPP (Austin et al., 2021) for code generation. To ensure fair comparisons, we evaluate against **SFT**, **TraceRL** (Wang et al., 2026b), and **TOPD** (Ren et al., 2026) as three primary baselines.

Implementation details. For mathematical reasoning, we follow TraceRL (Wang et al., 2026b) and use 8K math problems for post-training. For code generation, we use a curated subset of over 10,000 problems from AceCoder (Zeng et al., 2025). Student trajectories are collected using dynamic decoding with a batch size of 64 prompts (one rollout per prompt). We adopt the dynamic decoding schedule with confidence threshold 0.9 and temperature 1.0 for evaluation unless otherwise specified. We report mean accuracy over four sampled responses per problem (avg @4), except

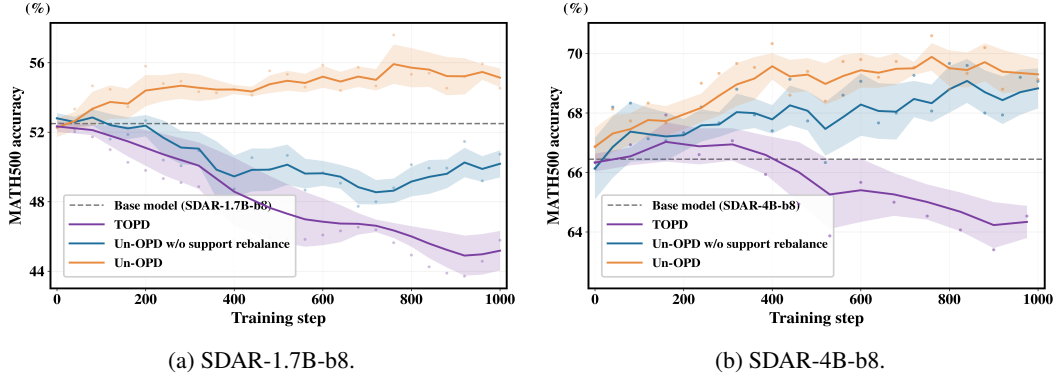


Fig. 3: **Training dynamics on MATH500.** We compare the base model (dashed line), TOPD, Un-OPD without support rebalance, and the complete Un-OPD. Solid lines show the smoothed performance trend.

for AMC 2023, where we use 16 responses per problem (avg @16). All models are trained for 1,000 gradient steps on an 8-GPU NVIDIA H20 server to guarantee convergence.

4.2 MAIN RESULTS

Tab. 1 shows distillation performance when scaling to students with a larger block size ($b = 8$) under a teacher ($b = 4$). Across both student scales, Un-OPD outperforms TOPD on all six benchmarks and achieves the highest or tied-highest average accuracy in both domains. For the SDAR-4B-b8 student, Un-OPD raises average accuracy in mathematical reasoning from 57.5% to 59.7% and in code generation from 58.7% to 63.4%. The advantage is even more striking on the smaller SDAR-1.7B-b8 student: while TOPD suffers from severe performance degradation and consistently falls below the base model, Un-OPD successfully stabilizes training, elevating math and code averages to 44.1% and 49.0%. Overall, these results support the effectiveness of addressing optimization biases when transferring knowledge into larger-block BDLMs. Moreover, for BDLMs with the same block size (*i.e.*, 4), Un-OPD still achieves better overall performance than TOPD. For example, the mathematical reasoning accuracy increases from 46.5% to 47.0% for SDAR-1.7B-b4, yielding a 0.5% gain. These results demonstrate that Un-OPD readily generalizes across varying block-size configurations.

Tab. 3: Sensitivity to the interpolation coefficient λ on MATH500.

λ	0.2	0.4	0.6	0.8	1.0
SDAR-1.7B-b8	53.7	54.9	54.1	52.4	50.9
SDAR-4B-b8	70.0	70.1	69.8	68.7	67.1

4.3 ABLATION STUDY

Ablation of core components. Tab. 2 ablates our proposed components on both students (SDAR-1.7B-b8 and SDAR-4B-b8). Directly applying standard OPD (TOPD) yields marginal gains on SDAR-4B-b8 and even causes performance degradation on the smaller SDAR-1.7B-b8, confirming the deficiency of trace-aligned distillation. Introducing **context-aligned distillation** improves all three reported benchmarks on both student scales, with MATH500 gains of 2.4% for the SDAR-4B-b8. However, context alignment alone leaves the weaker SDAR-1.7B-b8 student underperforming. Adding **support-rebalanced distillation** to calibrate updates at high-support positions improves accuracy on all three reported benchmarks for the SDAR-1.7B-b8 student (*e.g.*, a gain of 2.9% on MATH500). Finally, the **rollout reuse** mechanism reduces training overhead (from 152 to 72 GPU hours for 4B) while largely preserving accuracy. These ablations validate that each component is indispensable to achieving both superior accuracy and training efficiency.

Furthermore, Fig. 3 compares the training dynamics on MATH500 across both student scales. While standard TOPD suffers from severe late-stage degradation, Un-OPD maintains steady improvements throughout training. Crucially, omitting support rebalance causes the smaller SDAR-1.7B-b8 student to collapse below base accuracy and limits the gains of SDAR-4B-b8. This confirms that

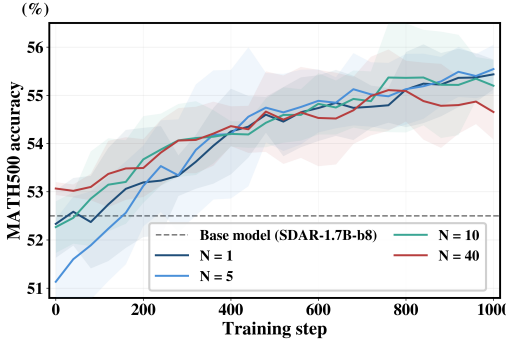


Fig. 4: Training dynamics on MATH500 with different rollout reuse epochs N on SDAR-1.7B-b8. Solid lines show accuracy trajectories; the dashed line denotes the base model.

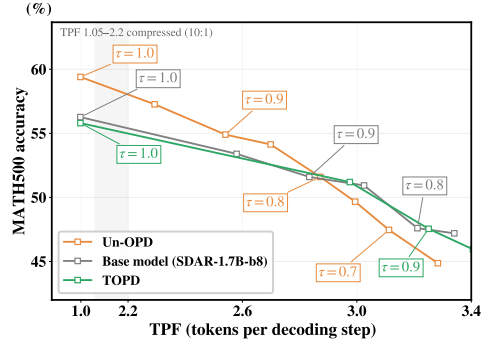


Fig. 5: Accuracy versus tokens per forward pass (TPF) on MATH500 for Un-OPD and the SDAR-1.7B-b8 base model across decoding thresholds τ .

support-rebalanced calibration is essential to ensure robust convergence, particularly for capacity-constrained models.

Hyper-parameter analysis of λ and N . We examine the sensitivity of Un-OPD to the interpolation coefficient λ and rollout reuse epoch number N on MATH500. For λ , Tab. 3 shows that both student scales achieve their highest reported accuracy at $\lambda = 0.4$: 54.9% for SDAR-1.7B-b8 and 70.1% for SDAR-4B-b8. Increasing λ beyond 0.4 reduces accuracy on both models, supporting a moderate calibration strength. For rollout reuse, Fig. 4 compares the training dynamics of SDAR-1.7B-b8 under $N \in \{1, 5, 10, 40\}$. Moderate reuse largely preserves performance: by the end of training, $N = 5$ and $N = 10$ achieve similar performance to $N = 1$ (the non-reused baseline). In contrast, with $N = 40$, performance declines during later training, likely due to off-policy shift from overly stale rollout trajectories. Taken together, we set $\lambda = 0.4$ and $N = 10$ to balance accuracy and rollout generation cost.

Analysis of tokens per forward pass (TPF). To evaluate the trade-off between reasoning accuracy and decoding parallelism, we sweep the dynamic decoding threshold τ on MATH500 (Fig. 5). Across high-to-moderate confidence thresholds, Un-OPD consistently establishes a superior Pareto frontier over the base model, achieving up to a +3.3% accuracy gain under conservative decoding ($\tau = 1.0$). However, as decoding becomes overly aggressive (higher TPF), this performance advantage narrows and eventually diminishes, leaving room for further improvement.

Out-of-distribution (OOD) transferability. To test cross-domain robustness, we evaluate SDAR-4B-b8 across math and code domains (Tab. 4). After post-training on math, Un-OPD largely preserves code-generation performance, maintaining virtually identical accuracy on MBPP (53.3%) and HumanEval (68.0%). Furthermore, code-trained Un-OPD exhibits modest positive transfer to math reasoning, outperforming Base by +0.9% on MATH500 and +1.1% on AMC2023. These results confirm that Un-OPD avoids out-of-domain degradation while facilitating cross-task generalization.

Tab. 4: Cross-domain accuracy of SDAR-4B-b8. Arrows denote training $\rightarrow$ evaluation domains.

Method	Code $\rightarrow$ Math		Math $\rightarrow$ Code	
	MATH500	AMC2023	MBPP	HumanEval
Base	66.3	42.5	53.5	67.9
Un-OPD	67.2	43.6	53.3	68.0

5 CONCLUSION

In this work, we pioneer the study of scaling on-policy distillation (OPD) to block diffusion language models (BDLMs) with varying block sizes. We uncover two fundamental optimization bottlenecks: *context misalignment* induced by mismatched block boundaries, and *overconfidence collapse* triggered by biased learning dynamics under reverse-KL supervision. To overcome these challenges, we introduce **Un-OPD**, which eliminates supervisory distortions via boundary-aware step filtering and

recalibrates optimization intensity across token supports. In addition, we integrate a rollout reuse mechanism that roughly halves training overhead with negligible accuracy loss. Comprehensive experiments on mathematical reasoning and code generation demonstrate that Un-OPD achieves remarkable stability and superior accuracy as well as substantial efficiency. Despite these gains, the effectiveness of Un-OPD may depend on the quality and suitability of the teacher’s supervision. Furthermore, pushing this Pareto frontier toward ultra-high parallelism regimes presents a compelling avenue for future research.

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